

PRAKTIKUM SISTEM OPERASI

MODUL VI

PROSES SEKUENSIAL DAN KONKUREN



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

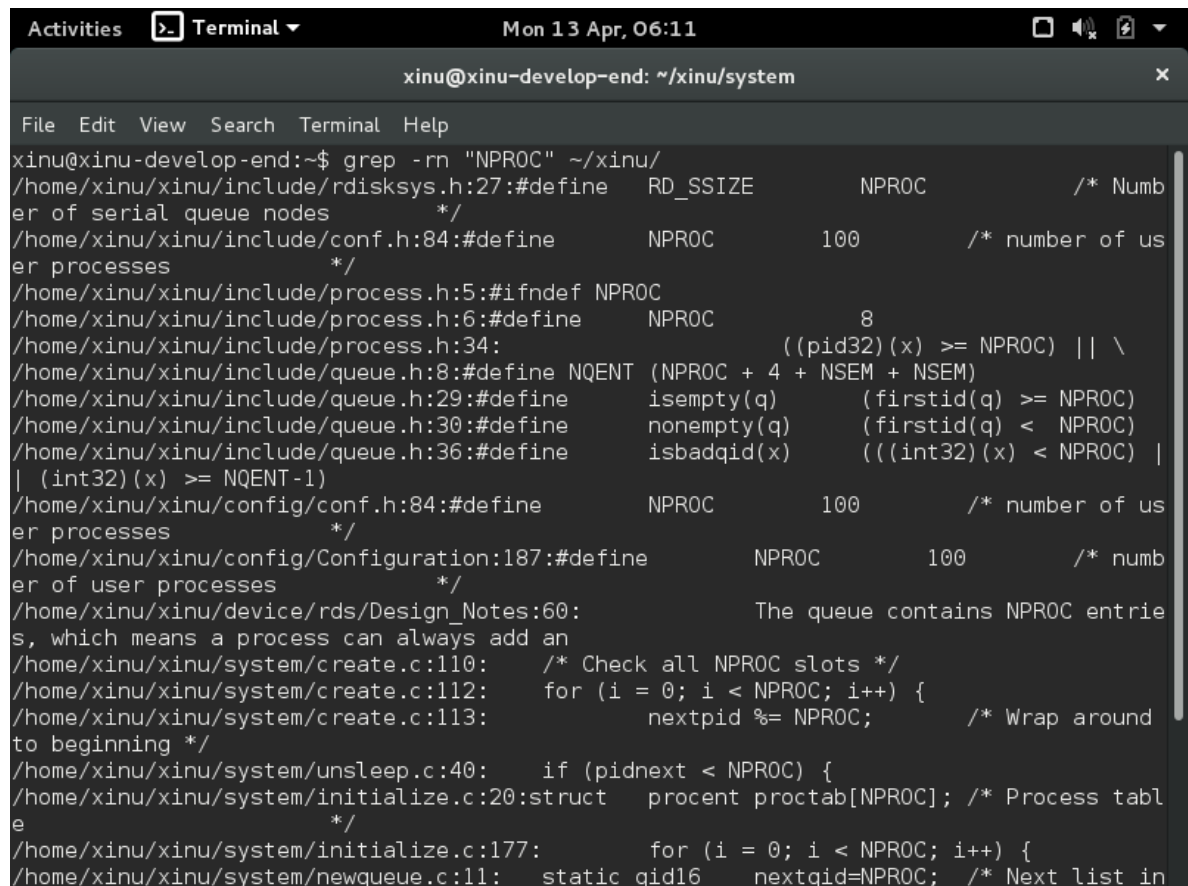
UNIVERSITAS TELKOM

2026

JURNAL

1. Konfigurasi Batas Proses: Cari di source code Xinu bagian yang mengatur jumlah maksimal proses. Sebutkan nilai defaultnya, lalu ubah batas tersebut menjadi 150.

Jawaban:



```
xinu@xinu-develop-end: ~/xinu/system
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ grep -rn "NPROC" ~/xinu/
/home/xinu/xinu/include/rdisksys.h:27:#define RD_SSIZE NPROC /* Numb
er of serial queue nodes */
/home/xinu/xinu/include/conf.h:84:#define NPROC 100 /* number of us
er processes */
/home/xinu/xinu/include/process.h:5:#ifndef NPROC
/home/xinu/xinu/include/process.h:6:#define NPROC 8
/home/xinu/xinu/include/process.h:34: ((pid32)(x) >= NPROC) || \
/home/xinu/xinu/include/queue.h:8:#define NQENT (NPROC + 4 + NSEM + NSEM)
/home/xinu/xinu/include/queue.h:29:#define isempty(q) (firstid(q) >= NPROC)
/home/xinu/xinu/include/queue.h:30:#define nonempty(q) (firstid(q) < NPROC)
/home/xinu/xinu/include/queue.h:36:#define isbadqid(x) (((int32)(x) < NPROC) |
| (int32)(x) >= NQENT-1)
/home/xinu/xinu/config/conf.h:84:#define NPROC 100 /* number of us
er processes */
/home/xinu/xinu/config/Configuration:187:#define NPROC 100 /* numb
er of user processes */
/home/xinu/xinu/device/rds/Design_Notes:60: The queue contains NPROC entrie
s, which means a process can always add an
/home/xinu/xinu/system/create.c:110: /* Check all NPROC slots */
/home/xinu/xinu/system/create.c:112: for (i = 0; i < NPROC; i++) {
/home/xinu/xinu/system/create.c:113: nextpid %= NPROC; /* Wrap around
to beginning */
/home/xinu/xinu/system/unsleep.c:40: if (pidnext < NPROC) {
/home/xinu/xinu/system/initialize.c:20:struct procent proctab[NPROC]; /* Process tabl
e */
/home/xinu/xinu/system/initialize.c:177: for (i = 0; i < NPROC; i++) {
/home/xinu/xinu/system/newqueue.c:11: static qid16 nextqid=NPROC; /* Next list in
```

```
Activities Terminal Mon 13 Apr, 06:10
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
GNU nano 2.2.6 File: process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 150
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */

[ Read 62 lines ]
^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

```
Activities Terminal Mon 13 Apr, 07:34
xinu@xinu-develop-end: ~/xinu/include

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd ~/xinu/include
xinu@xinu-develop-end:~/xinu/include$ cat process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      150
#endif

/* Process state constants */

#define PR_FREE      0      /* Process table entry is unused      */
#define PR_CURR      1      /* Process is currently running      */
#define PR_READY     2      /* Process is on ready queue      */
#define PR_RECV      3      /* Process waiting for message      */
#define PR_SLEEP     4      /* Process is sleeping      */
#define PR_SUSP      5      /* Process is suspended      */
#define PR_WAIT      6      /* Process is on semaphore queue      */
#define PR_RECTIM    7      /* Process is receiving with timeout */

/* Miscellaneous process definitions */

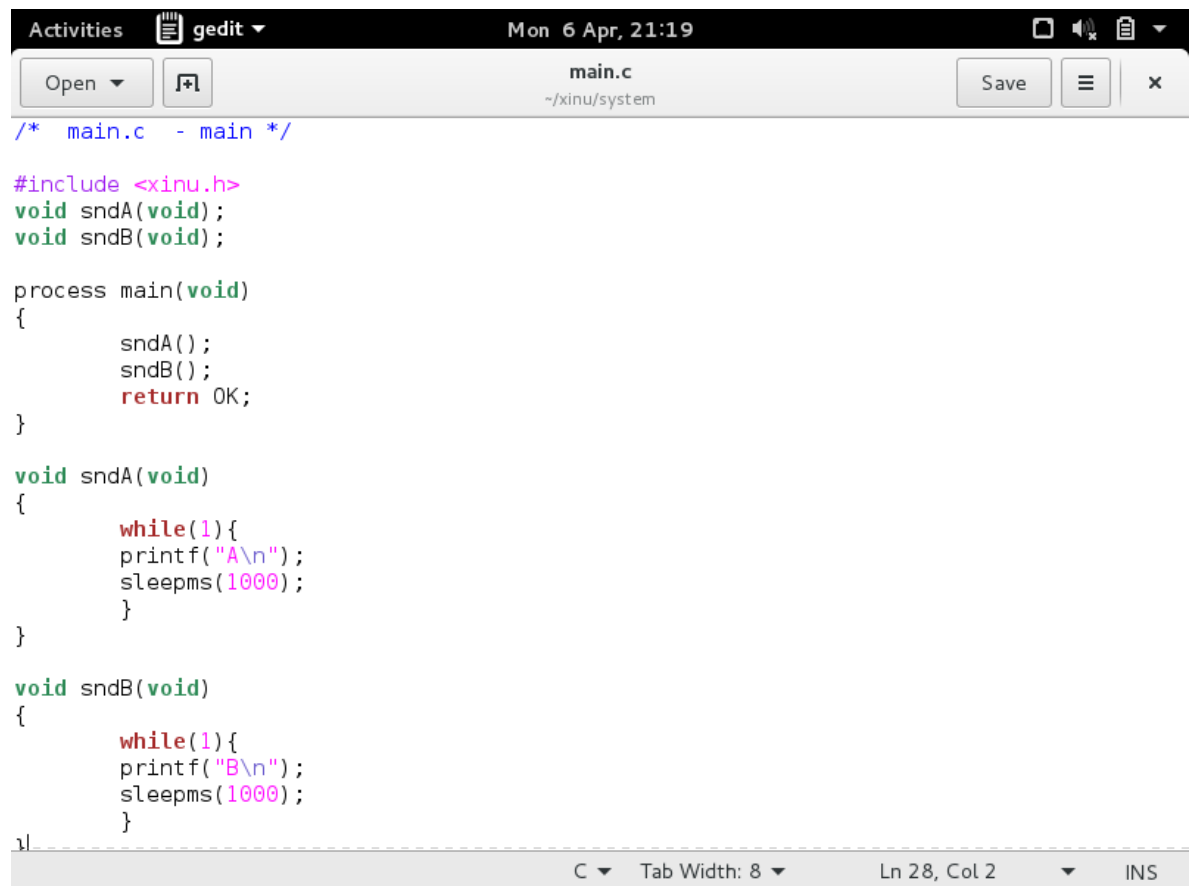
#define PNMLEN      16      /* Length of process "name"      */
#define NULLPROC    0      /* ID of the null process      */

/* Process initialization constants */
```

- a. Cari file process.h dengan perintah: **grep -rn "NPROC" ~/xinu/**
- b. Masuk ke folder include: **cd ~/xinu/include**
- c. Edit file process.h: **nano process.h**
- d. Cari baris **#define NPROC 8**, ubah angka 8 → 150
Simpan: **Ctrl+O** → **Enter** → **Ctrl+X**
- e. Verifikasi:
- f. **cat process.h**

2. Run: main_ex1

Jawaban:



```
/* main.c - main */

#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    sndA();
    sndB();
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}
```

```
Activities Terminal Mon 6 Apr, 21:41
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Xinu for Vbox -- version #101 (xinu) Tue 7 Apr 00:22:23 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
  4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
  115231 bytes of Xinu code.
    [0x00100000 to 0x0011C21E]
  148456 bytes of data.
    [0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
A
A
A
A
A

```

Urutan pengerjaan:

1. Masuk ke folder system: `cd ~/xinu/system`
2. Edit main.c: `gedit main.c`
3. Isi kode:

```
/* main.c - main */
#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void) {
    sndA();
    sndB();
    return OK;
}

void sndA(void) {
    while(1) {
        printf("A\n");
        sleepms(1000);
    }
}
```

```
    }  
}  
void sndB(void) {  
    while(1) {  
        printf("B\n");  
        sleepms(1000);  
    }  
}
```

4. **Simpan, lalu compile:** `cd ~/xinu/compile && make`
5. **Jalankan minicom:** `sudo minicom`

3. Run: main_ex2

Jawaban:



```
/* main.c - main */

#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    resume(create(sndA, 1024, 20, "process A", 0));
    resume(create(sndB, 1024, 20, "process B", 0));
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}
```



```
Activities Terminal Mon 6 Apr, 22:19
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Xinu for Vbox -- version #102 (xinu) Tue 7 Apr 01:13:22 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
  4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
  115359 bytes of Xinu code.
    [0x00100000 to 0x0011C29E]
  148456 bytes of data.
    [0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Urutan pengerjaan:

1. Edit kembali ~/xinu/system/main.c:

```
/* main.c - main */
#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void) {
    resume(create(sndA, 1024, 20, "process A", 0));
    resume(create(sndB, 1024, 20, "process B", 0));
    return OK;
}

void sndA(void) {
    while(1) {
        printf("A\n");
        sleepms(1000);
    }
}
```

```
void sndB(void) {  
    while(1) {  
        printf("B\n");  
        sleepms(1000);  
    }  
}
```

2. Simpan → compile → jalankan minicom

4. Run: main_ex3

Jawaban:



```
/* main.c - main */

#include <xinu.h>
void sndChar(char);

process main(void)
{
    resume(create(sndChar, 1024, 20, "send A", 1, 'A'));
    resume(create(sndChar, 1024, 20, "send B", 1, 'B'));
    return OK;
}

void sndChar(char c)
{
    while(1){
        printf("%c \n", c);
        sleepms(1000);
    }
}
```

C ▾ Tab Width: 8 ▾ Ln 19, Col 2 ▾ INS

```
Activities Terminal Mon 13 Apr, 06:17
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu/include$ nano process.h
xinu@xinu-develop-end:~/xinu/include$ cd ~/xinu/system
xinu@xinu-develop-end:~/xinu/system$ ls
ascdate.c      freebuf.c      init.c          open.c          ready.c         sleep.c
bufinit.c      freemem.c      initialize.c    panic.c         receive.c       stacktrace.c
chprio.c       getbuf.c       insert.c        pci.c           recvclr.c       start.S
clkdisp.S      getc.c         insertd.c       ptclear.c       recvtime.c      suspend.c
clkhandler.c   getdev.c       intr.S          ptcount.c       resched.c       unsleep.c
clkinit.c      getitem.c      ioerr.c         ptcreate.c      resume.c        userret.c
close.c        getmem.c       ionull.c        ptdelete.c      seek.c          wait.c
conf.c         getpid.c       kill.c          ptinit.c        semcount.c      wakeup.c
control.c      getprio.c      kprintf.c       ptrecv.c        semcreate.c     write.c
create.c       getstk.c       main.c          ptreset.c       semdelete.c     xdone.c
ctxsw.S        getticks.c     mark.c          ptsend.c        semreset.c      yield.c
debug.c        gettime.c      meminit.c       putc.c          send.c
evec.c         getutime.c     mkbufpool.c     queue.c         signal.c
exit.c         i386.c         newqueue.c      read.c          signaln.c
xinu@xinu-develop-end:~/xinu/system$ gedit main.c
xinu@xinu-develop-end:~/xinu/system$ cd ~/xinu/compile && make
Rebuilding the .defs file
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/main.o ../system/main.c

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$ sudo minicom
```

```
Activities Terminal Mon 13 Apr, 06:18
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

Xinu for Vbox -- version #102 (xinu) Tue 7 Apr 01:13:22 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
  4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
  115327 bytes of Xinu code.
    [0x00100000 to 0x0011C27E]
  148456 bytes of data.
    [0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
A
B
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Urutan pengerjaan:

1. Edit main.c:

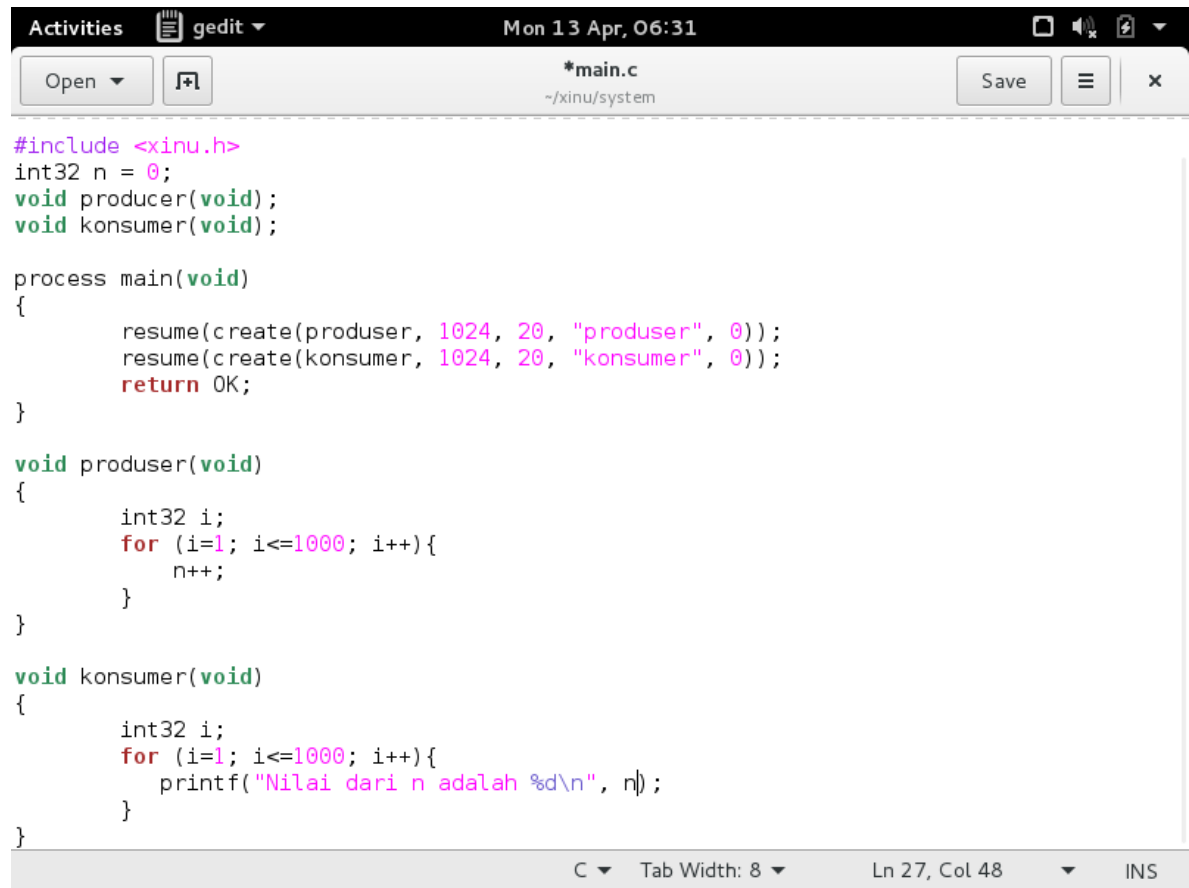
```
/* main.c - main */
#include <xinu.h>
void sndChar(char);

process main(void) {
    resume(create(sndChar, 1024, 20, "send A", 1, 'A'));
    resume(create(sndChar, 1024, 20, "send B", 1, 'B'));
    return OK;
}

void sndChar(char c) {
    while(1) {
        printf("%c\n", c);
        sleepms(1000);
    }
}
```

2. Simpan → compile → jalankan minicom
5. Analisis Sinkronisasi: Buat proses Producer (isi variabel n 1-1000) dan Consumer (cetak variabel n). Jelaskan mengapa outputnya tidak sesuai urutan/ekspektasi (analisis race condition).

Jawaban:



```
#include <xinu.h>
int32 n = 0;
void producer(void);
void konsumer(void);

process main(void)
{
    resume(create(produser, 1024, 20, "produser", 0));
    resume(create(konsumer, 1024, 20, "konsumer", 0));
    return OK;
}

void produser(void)
{
    int32 i;
    for (i=1; i<=1000; i++){
        n++;
    }
}

void konsumer(void)
{
    int32 i;
    for (i=1; i<=1000; i++){
        printf("Nilai dari n adalah %d\n", n);
    }
}
```

C Tab Width: 8 Ln 27, Col 48 INS

```
Activities Terminal Mon 13 Apr, 06:32
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd ~/xinu/system
xinu@xinu-develop-end:~/xinu/system$ gedit main.c

(gedit:1194): GLib-GObject-WARNING **: The property GtkTextTag:foreground-gdk is deprecated and shouldn't be used anymore. It will be removed in a future version.
xinu@xinu-develop-end:~/xinu/system$ cd ~/xinu/compile && make
Rebuilding the .defs file
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall -O0 -DBRELOC=0x150000 -DBOOTPLC=0x150000 -DBSDURG -DVERSION=\"\"`cat version`\" -I../include -o binaries/main.o ../system/main.c
../system/main.c: In function 'main':
../system/main.c:10:16: error: 'produser' undeclared (first use in this function)
  resume(create(produser, 1024, 20, "produser", 0));
                ^
../system/main.c:10:16: note: each undeclared identifier is reported only once for each function it appears in
.defs:328: recipe for target 'binaries/main.o' failed
make: *** [binaries/main.o] Error 1
xinu@xinu-develop-end:~/xinu/compile$ sudo minicom
```

```
Activities Terminal Mon 13 Apr, 06:32
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
Xinu for Vbox -- version #102 (xinu) Tue 7 Apr 01:13:22 EDT 2026
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
115327 bytes of Xinu code.
    [0x00100000 to 0x0011C27E]
148456 bytes of data.
    [0x0011FEE0 to 0x001442C7]
Obtained IP address 10.0.3.15 (0xa00030f)
A
B
A
B
A
B
A
B
A
B
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Urutan pengerjaan:

1. Edit main.c:

```

/* main.c - main */
/* Nama: [nama kamu] - NIM: [nim kamu] */
#include <xinu.h>

int32 n = 0;

void produser(void);
void konsumer(void);

process main(void) {
    resume(create(produser, 1024, 20, "produser", 0));
    resume(create(konsumer, 1024, 20, "konsumer", 0));
    return OK;
}

void produser(void) {
    int32 i;
    for (i=1; i<=1000; i++) {
        n++;
    }
}

```



```

    }
}
void konsumer(void) {
    int32 i;
    for (i=1; i<=1000; i++) {
        printf("Nilai dari n adalah %d\n", n);
    }
}

```

2. Simpan → compile → jalankan minicom

3. Screenshot output

4. Tulis penjelasan: Kenapa hasilnya langsung 1000 semua?

Karena kedua proses berjalan konkuren, ada kemungkinan proses produser selesai duluan melakukan loop 1000 kali (sehingga $n = 1000$) sebelum konsumer sempat membaca nilai n yang kecil. Akibatnya, saat konsumer mulai membaca, nilai n sudah = 1000 dan semua output menampilkan 1000. Inilah yang disebut race condition — urutan eksekusi proses tidak bisa diprediksi tanpa sinkronisasi.